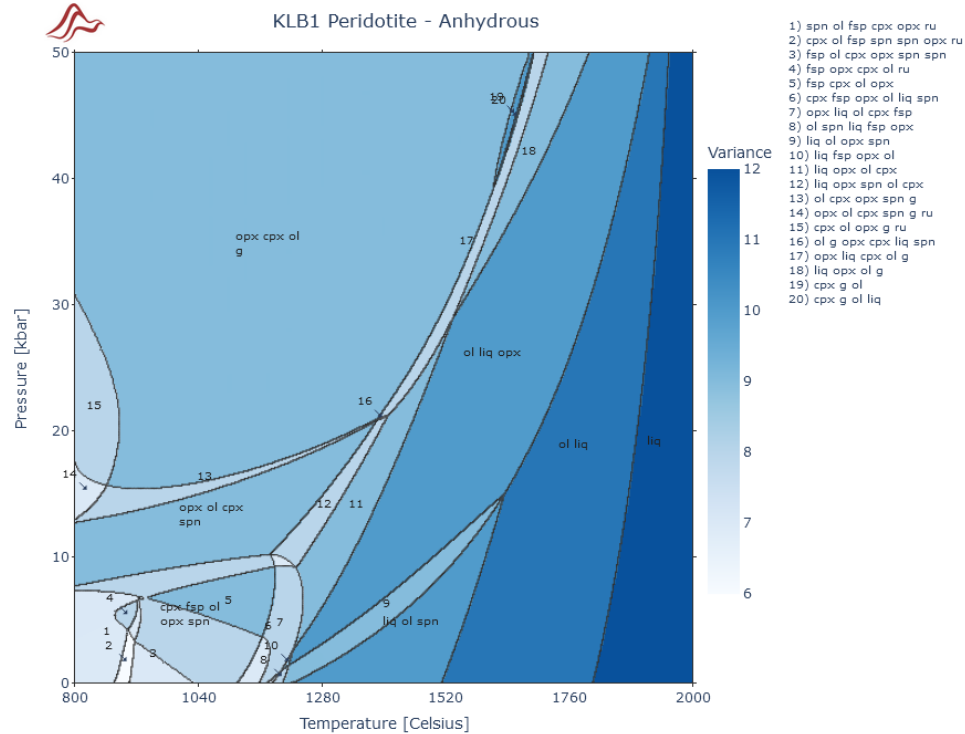


MAGEMinApp: introduction



Main contributors

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J-F. Moyen
S. Schorn
H. Dominguez
P. Lanari
J. Forshaw
T. Holland



MAGEMin github: ComputationalThermodynamics

- First, access the online documentation:



https://computationalthermodynamics.github.io/MAGEMin_C.jl/dev/

MAGEMin Toolset
Phase equilibrium modelling

A set of tools to compute phase equilibria for Earth-like compositions

[Getting Started](#) [MAGEMin_C.jl API](#) [View on GitHub](#)

MAGEMinApp
Effortlessly compute phase diagrams and Pressure-Temperature paths

MAGEMin_C
Set of function to freely compute phase equilibrium in Julia

MAGEMin
C program backend of MAGEMin

What is MAGEMin?

Home

Method

MAGEMinApp.jl

- Introduction
- Installation
- Bulk-rock input file
- Phase diagrams tutorials
- P-T-X paths tutorials
- Citation
- Interface

MAGEMin_C.jl

- Introduction
- Installation
- Tutorials
- API

MAGEMin

- Introduction
- Compilation
- Tutorials

MAGEMinApp: launching the App in parallel

VISUAL STUDIO CODE / TERMINAL MAC, LINUX

```
seph42@DESKTOP-2V82075:~$ julia -t 6

Documentation: https://docs.julialang.org
Type "?" for help, "]" for Pkg help.

Version 1.10.0 (2023-12-25)
Official https://julialang.org/ release

julia> using MAGEMinApp
App()
Using libMAGEMin.dylib from MAGEMin_jll

julia> App()
[ Info: Listening on: 127.0.0.1:8050, thread id: 2
```

```
julia -t 6
using MAGEMinApp
App()
```

POWERSHELL WINDOWS

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Seph> $env:JULIA_NUM_THREADS=6
PS C:\Users\Seph> julia

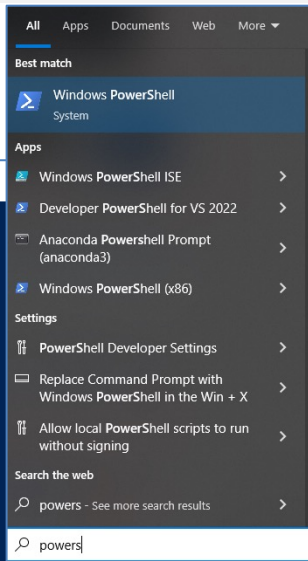
Documentation: https://docs.julialang.org
Type "?" for help, "]" for Pkg help.

Version 1.9.3 (2023-08-24)
Official https://julialang.org/ release

julia> using MAGEMinApp
Using libMAGEMin.dylib from MAGEMin_jll

julia> App()
[ Info: Listening on: 127.0.0.1:8050, thread id: 1
```

```
$env:JULIA_NUM_THREADS=6
julia
using MAGEMinApp
App()
```



MAGEMinApp: interface (Phase diagrams)



Reminder: Don't forget to properly cite the references used to create the diagrams by using the 'export references' buttons!

Make sure you use the last version of the app 'add MAGEMinApp@0.8.6' (Julia v1.10+)



Phase diagramsPTX pathIsentropic pathGeneral information

SetupDiagramTrace-elements

Phase diagram parameters

Thermodynamic database

Igneous (Green et al., 2025, after H18)

Dataset

ds636

Phase selection

Solution phase

spl

bi

cd

epx

ep

g

amp

llm

liq

ol

opx

fsp

fl

mu

fper

Pure phase

ne

q

crst

ard

coe

stv

ty

sill

and

ru

sph

H2O

Diagram type

P-T diagram (fixed composition)

Solidus H₂O-saturated

false

TE predictive model

false

Pressure [kbar]

min

0.01

max

20.01

Temperature [°C]

min

800

max

1400

Bulk-rock composition

mol%

Drag and drop or select bulk-rock file

KLB1 Peridotite - Anhydrous

oxide	fraction
SiO ₂	38.494
Al ₂ O ₃	1.776
CaO	2.824
MgO	50.566
FeO	5.886
K ₂ O	0.01
Na ₂ O	0.25
TiO ₂	0.1
O	0.096
Cr ₂ O ₃	0.109
H ₂ O	0

ig

General parameters

Title

KLB1 Peridotite - Anhydrous

UpdateReset

Compute phase diagram

Save/Load Diagram

filename

Save state

Load state

State/CSV directory: /Users/nicolasrieli/seph/MAGEMinApp_jl_v0.8.7

Help and contact

Links

[Tutorials](#)[Post issue](#)[Open discussion](#)

Comments

Set of example on how to use MAGEMinApp
Something is not working properly?
Need additional options?

Contributors

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MAGEMinApp: interface (Phase diagrams)

Phase diagramsPTX pathIsentropic pathGeneral information

SetupDiagramTrace-elements

Phase diagram parameters

Thermodynamic database

Igneous (Green et al., 2025, after H18)

Dataset

ds636

Phase selection

Solution phase

☒ spl

☒ bi

☒ cd

☒ cpx

☒ ep

☒ g

☒ amp

☒ ilm

☒ liq

☒ ol

☒ opx

☒ fsp

☒ fl

☒ mu

☒ fper

Pure phase

☒ ne

☒ q

☒ crst

☒ ard

☒ coe

☒ stv

☒ ky

☒ sill

☒ and

☒ ru

☒ sph

☒ H2O

Diagram type

P-T diagram (fixed composition)

Solidus H₂O-saturated

false

TE predictive model

false

Pressure [kbar]

min

0.01

max

20.01

Bulk-rock composition

mol%

Drag and drop or select bulk-rock file

KLB1 Peridotite - Anhydrous

oxide	fraction
SiO ₂	38.494
Al ₂ O ₃	1.776
CaO	2.824
MgO	50.566
FeO	5.886
K ₂ O	0.01
Na ₂ O	0.25
TiO ₂	0.1
O	0.096
Cr ₂ O ₃	0.109
H ₂ O	0

General parameters

Title

KLB1 Peridotite - Anhydrous

UpdateReset

Compute phase diagram

Save/Load Diagram

filename

Save stateLoad state

State/CSV directory: /Users/nicolasri/seph/MAGEMinApp_jl_v0.8.7

Help and contact

Links

[Tutorials](#)[Post issue](#)[Open discussion](#)

Comments

Set of example on how to use MAGEMinApp
Something is not working properly?
Need additional options?

Contributors

Grid properties

Bulk rock composition(s)

- Save/load state
- Compute diagram

MAGEMinApp: Thermodynamic database

Phase diagram parameters

Thermodynamic database

Solution phase selection

Diagram type

TE predictive model

Pressure [kbar]

Temperature [°C]

Initial grid subdivision

Refinement type

Refinement levels

Buffer

Solver

Verbose

Specific Cp

Igneous (Holland et al., 2018)

Metapelite (White et al., 2014)

Metabasite (Green et al., 2016)

Igneous (Holland et al., 2018)

Ultramafic (Evans & Frost, 2021)

min max

0,01 20,01

min max

800 1400

3

Phases only

2

no buffer

Hybrid

none

G0

id=6a7eedc6-5f5b-4ca6-a39d-d968df521e13, MAGEMinApp GUI v=0.4.0

- Mantle (Holland et al., 2013)
NCFMAS
- Metapelite (White et al., 2014)
MnNCKFMASHTO
- Metabasite (Green et al., 2016)
NCKFMASHTO
- Igneous (Green et al., 2025 after Holland et al., 2018)
NCKFMASHTOCr
- Ultramafic (Evans & Frost, 2021)
FMASOSH
- Dry igneous alkaline (Weller et al., in 2024)
NCKFMASHTOCr
- Mantle (Stixrude and Lithgow-Bertelloni, 2011, 2021)
NCFMAS (upcoming)

MAGEMinApp: solution phase selection

Thermodynamic database

Dataset

Phase selection

Solution phase

Pure phase

Diagram type

Solidus H₂O-saturated

TE predictive model

Pressure [kbar]

Temperature [°C]

Igneous (Green et al., 2025, after H18)

ds636

spl
bi
cd
cpx
ep
g
amp
ilm
liq
ol
opx
fsp
fl
mu
fper

ne
q
crst
trd
coe
stv
ky
sill
and
ru
sph
H2O

P-T diagram (fixed composition)

false

false

min max

0.01 20.01

min max

800 1400

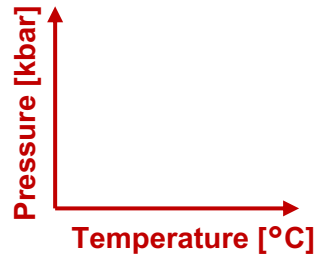
- Allow to deactivate phase
- Note that solution phase models from different dataset cannot be mixed (yet)

MAGEMinApp: diagram types

Phase diagram parameters

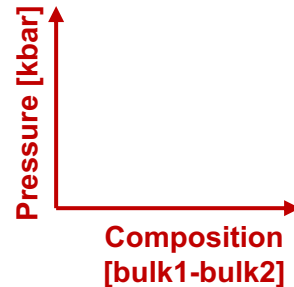
Thermodynamic database	Igneous (Green et al., 2025, after H18)	
Dataset	ds636	
Phase selection	☐ Solution phase ☐ Pure phase	
Diagram type	P-T diagram (fixed composition)	
Solidus H ₂ O-saturated	false	
TE predictive model	false	
Pressure [kbar]	min: 0.01	max: 20.01
Temperature [°C]	min: 400	max: 1000
Initial grid subdivision	4	16 × 16 grid
Refinement type	Phases only	
Refinement levels	4	
Boost mode	true	
Buffer	no buffer	
Solver	Legacy	
Verbose	none	
Specific Cp	G0	

P-T



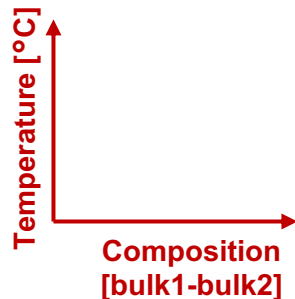
P-X

Fixed temperature

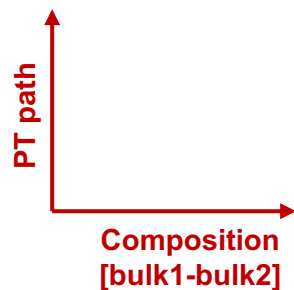


T-X

Fixed pressure



PT-X



MAGEMinApp: diagram types

Phase diagram parameters

Thermodynamic database	Igneous (Green et al., 2025, after H18) ▼	
Dataset	ds636 ▼	
Phase selection	☐ Solution phase ☐ Pure phase	
Diagram type	P-T diagram (fixed composition) ▼	
Solidus H ₂ O-saturated	false ▼	

TE predictive model	false ▼	
---------------------	---------	--

Pressure [kbar]	min	max
	0.01 ▼	20.01 ▼
Temperature [°C]	min	max
	400 ▼	1000 ▼

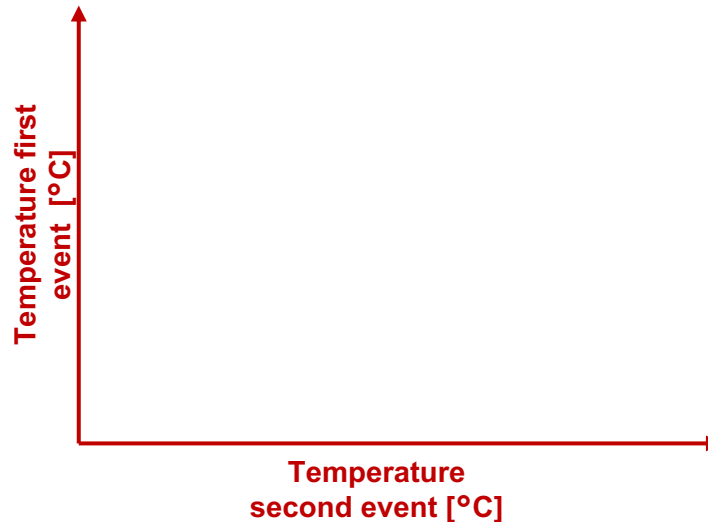
Initial grid subdivision	4 ▼	16 × 16 grid
Refinement type	Phases only ▼	
Refinement levels	4 ▼	
Boost mode	true ▼	

Buffer	no buffer ▼	
Solver	Legacy ▼	
Verbose	none ▼	
Specific Cp	G0 ▼	

T-T

**Fixed
pressure**

Poly-metamorphic diagram



MAGEMinApp: Trace element predictive model

Implemented with the help of J-F Moyen (LMV) and H. Dominguez (Uni - Bern)

Phase diagram parameters

Thermodynamic database: Igneous (Green et al., 2025, after H18)

Dataset: ds636

Phase selection: ☐ Solution phase ☐ Pure phase

Diagram type: P-T diagram (fixed composition)

Solidus H₂O-saturated: false

TE predictive model: false

Pressure [kbar]: min 0.01 max 20.01

Temperature [°C]: min 400 max 1000

Initial grid subdivision: 4 16 × 16 grid

Refinement type: Phases only

Refinement levels: 4

Boost mode: true

Buffer: no buffer

Solver: Legacy

Verbose: none

Specific Cp: G0

- Activate trace-element partitioning at supra-solidus conditions
- Displays trace element bulk panel on the right
- Gives the option to select among 3 zirconium saturation models

Watson & Harrison (1983)

Boehnke et al. (2013)

Crisp and Berry (2022)

General parameters

Title: KLB1 Peridotite - Anhydrous

State/CSV directory: /home/seph42/MAGEMinApp/ij_v04.0

Trace-element composition

Drag and drop or select trace-element file

Pyrolite (Primitive Mantle) ▾

elements	µg/g
Rb	0.6
Ba	6.6
Th	0.0795
U	0.0203
Nb	0.658
Ta	0.037
La	0.648
Ce	1.675
Pb	0.15
Pr	0.254
Sr	19.9
Nd	1.25

MAGEMinApp: grid subdivision and refinement

Phase diagram parameters

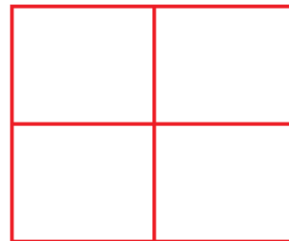
Thermodynamic database	Igneous (Green et al., 2025, after H18) ▼
Dataset	ds636 ▼
Phase selection	☐ Solution phase ☐ Pure phase
Diagram type	P-T diagram (fixed composition) ▼
Solidus H ₂ O-saturated	false ▼

TE predictive model	false ▼
---------------------	---------

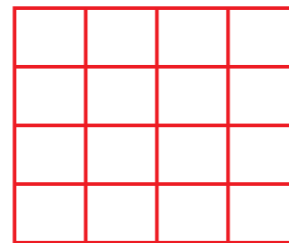
	min	max
Pressure [kbar]	0.01 ▼	20.01 ▼
	min	max
Temperature [°C]	400 ▼	1000 ▼

Initial grid subdivision	4 ▼	16 × 16 grid
Refinement type	Phases only ▼	
Refinement levels	4 ▼	
Boost mode	true ▼	

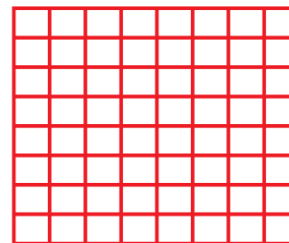
Buffer	no buffer ▼
Solver	Legacy ▼
Verbose	none ▼
Specific Cp	G0 ▼



1 initial grid subdivisions



2 initial grid subdivisions



3 initial grid subdivisions

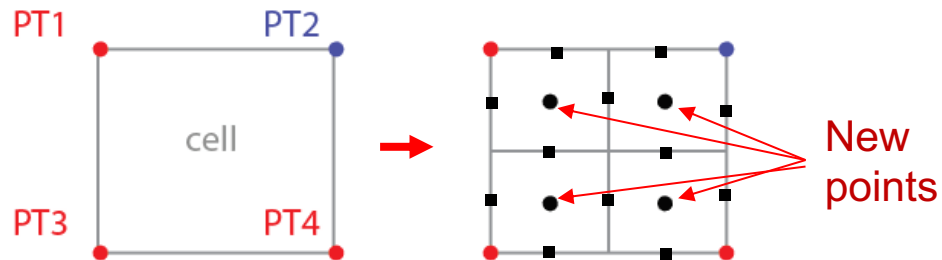
**Number of cells
 $(2^{n_subdivision}-1)^2$**

MAGEMinApp: grid subdivision and refinement

Phase diagram parameters

Thermodynamic database	Igneous (Green et al., 2025, after H18)
Dataset	ds636
Phase selection	☐ Solution phase ☐ Pure phase
Diagram type	P-T diagram (fixed composition)
Solidus H ₂ O-saturated	false
TE predictive model	false
Pressure [kbar]	min: 0.01 max: 20.01
Temperature [°C]	min: 400 max: 1000
Initial grid subdivision	4 16 x 16 grid
Refinement type	Phases only
Refinement levels	4
Boost mode	true
Buffer	no buffer
Solver	Legacy
Verbose	none
Specific Cp	G0

- **Phase only**
refines (split) cells when a phase change is recorded in one of the four corners
- **Dominant end-member**
refines (split) cells when the dominant end-member of any solution phase model changes in one of the four corners



- **Refinement levels**
Number of successive grid refinements

MAGEMinApp: grid refinement

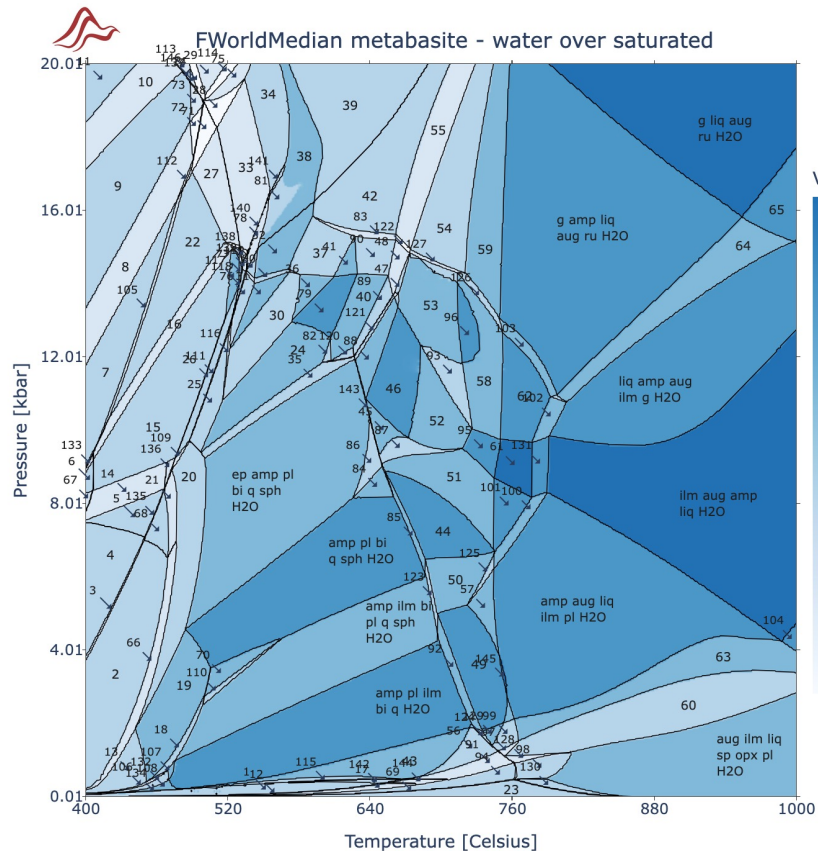
Export q for LaMEM

Export for GeoModel



Refine uniformly

Refine phase boundaries



- 1) act pl bi act sp ilm q H2O
- 2) pl bi chl act ep q sph H2O
- 3) act bi chl ep abc pl q sph H2O
- 4) bi abc ep chl act q sph H2O
- 5) abc bi chl tr ep q sph H2O
- 6) gl chl bi act ep q sph H2O
- 7) mu ep chl gl aug q sph H2O
- 8) mu gl aug chl ep q law sph H2O
- 9) gl chl mu aug q law sph H2O
- 10) chl gl aug mu tr q law sph H2O
- 11) chl gl mu tr q law sph H2O
- 12) bi pl act opx ilm sp q H2O
- 13) amp pl chl bi act q sph H2O
- 14) ep tr chl gl abc bi q sph H2O
- 15) ep tr chl bi gl q sph H2O
- 16) tr bi ep gl mu chl q sph H2O
- 17) ilm pl act sp bi aug opx q H2O
- 18) chl bi amp pl ilm q sph H2O
- 19) amp pl chl bi q sph H2O
- 20) ep bi amp chl pl q sph H2O
- 21) ep chl tr pl amp bi q sph H2O
- 22) gl ep tr mu chl q sph H2O
- 23) ilm pl opx afs aug sp q H2O
- 24) amp mu bi ep q sph H2O
- 25) amp bi chl ep mu q sph H2O
- 26) chl gl ep bi amp q sph H2O
- 27) gl chl g mu tr ep q sph H2O
- 28) g chl tr mu ep q law ru H2O
- 29) tr gl g aug mu q law ru H2O
- 30) bi amp mu ep q ru sph H2O
- 31) amp mu bi ep q ru H2O
- 32) amp ep mu g q ru H2O
- 33) tr mu g chl ep q ru H2O
- 34) mu g gl ep tr q ru H2O
- 35) amp mu ep bi pl q sph H2O
- 36) amp mu ep q ru sph H2O
- 37) amp mu g ep q ru sph H2O
- 38) gl g ep mu q ru H2O
- 39) ep mu gl g aug q ru H2O
- 40) mu ep aug amp q sph H2O
- 41) ep g amp mu q sph H2O
- 42) mu aug g amp ep q ru H2O
- 43) amp bi ilm nl so a H2O

MAGEMinApp: grid refinement

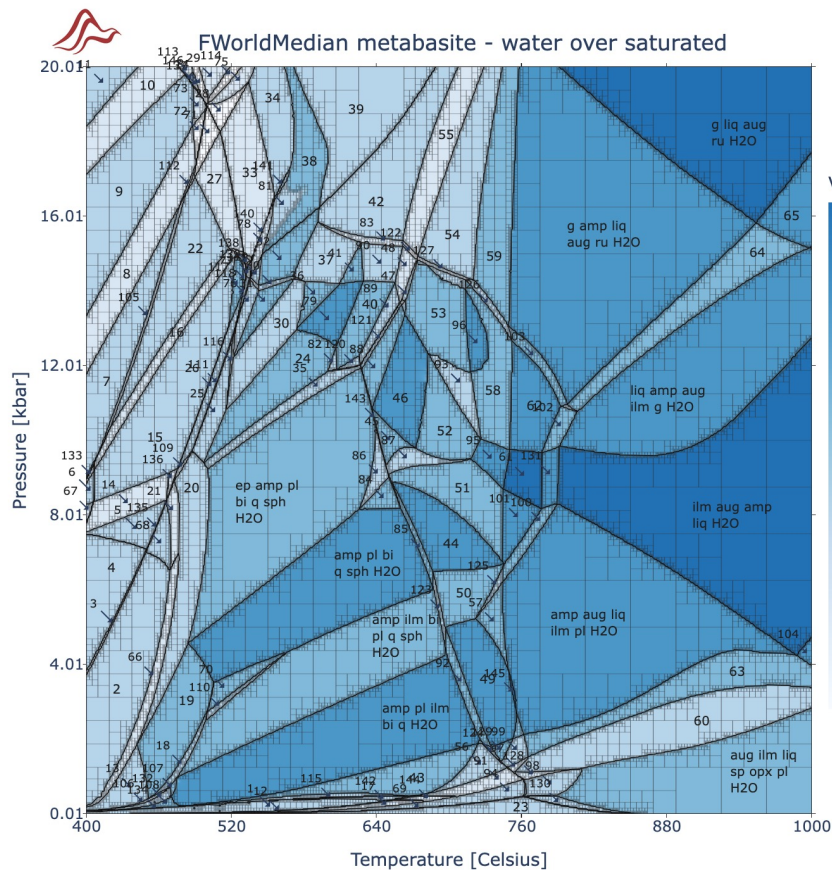
Export q for LaMEM

Export for GeoModel



Refine uniformly

Refine phase boundaries



- 1) act pl bi act sp ilm q H2O
- 2) pl bi chl act ep q sph H2O
- 3) act bi chl ep abc pl q sph H2O
- 4) bi abc ep chl act q sph H2O
- 5) abc bi chl tr ep q sph H2O
- 6) gl chl bi act ep q sph H2O
- 7) mu ep chl gl aug q sph H2O
- 8) mu gl aug chl ep q law sph H2O
- 9) gl chl mu aug q law sph H2O
- 10) chl gl aug mu tr q law sph H2O
- 11) chl gl mu tr q law sph H2O
- 12) bi pl act opx ilm sp q H2O
- 13) amp pl chl bi act q sph H2O
- 14) ep tr chl gl abc bi q sph H2O
- 15) ep tr chl bi gl q sph H2O
- 16) tr bi ep gl mu chl q sph H2O
- 17) ilm pl act sp bi aug opx q H2O
- 18) chl bi amp pl ilm q sph H2O
- 19) amp pl chl bi q sph H2O
- 20) ep bi amp chl pl q sph H2O
- 21) ep chl tr pl amp bi q sph H2O
- 22) gl ep tr mu chl q sph H2O
- 23) ilm pl opx afs aug sp q H2O
- 24) amp mu bi ep q sph H2O
- 25) amp bi chl ep mu q sph H2O
- 26) chl gl ep bi amp q sph H2O
- 27) gl chl g mu tr ep q sph H2O
- 28) g gl chl tr mu ep q law ru H2O
- 29) tr gl g aug mu q law ru H2O
- 30) bi amp mu ep q ru sph H2O
- 31) amp mu bi ep q ru H2O
- 32) amp ep mu g q ru H2O
- 33) tr mu g gl chl ep q ru H2O
- 34) mu g gl ep tr q ru H2O
- 35) amp mu ep bi pl q sph H2O
- 36) amp mu ep q ru sph H2O
- 37) amp mu g ep q ru sph H2O
- 38) gl g ep mu q ru H2O
- 39) ep mu gl g aug q ru H2O
- 40) mu ep aug amp q sph H2O
- 41) ep g amp mu q sph H2O
- 42) mu aug g amp ep q ru H2O
- 43) amo bi ilm ol so q H2O

MAGEMinApp: buffers

Phase diagram parameters

Thermodynamic database: Igneous (Green et al., 2025, after H18)

Dataset: ds636

Phase selection: ☐ Solution phase ☐ Pure phase

Diagram type: P-T diagram (fixed composition)

Solidus H₂O-saturated: false

TE predictive model: false

Pressure [kbar]: min 0.01 max 20.01

Temperature [°C]: min 400 max 1000

Initial grid subdivision: 4 16 × 16 grid

Refinement type: Phases only

Refinement levels: 4

Boost mode: true

Buffer: no buffer

Solver: Legacy

Verbose: none

Specific Cp: G0

- **Oxygen buffers**
QFM, MW, QIF, CCO, HM, NNO, IW
- **Fix system activity**
H₂O, O₂, SiO₂, FeO, MgO, Al₂O₃, TiO₂

Bulk-rock composition

Drag and drop or select bulk-rock file

KLB1 Peridotite - Anhydrous

oxide	mol_fraction
SiO ₂	38.494
Al ₂ O ₃	1.776
CaO	2.824
MgO	50.566
FeO	5.886
K ₂ O	0.01
Na ₂ O	0.25
TiO ₂	0.1
O	0.096
Cr ₂ O ₃	0.109
H ₂ O	0

Buffer multiplier: 0

- **Selecting a buffer displays the option to apply a shift in the RTlog scale**
e.g., +2 QFM
- **Allow to provide the activity value in case activity is selected**

MAGEMinApp: Specific heat capacity (latent heat of reaction)

Phase diagram parameters

Thermodynamic database	Igneous (Green et al., 2025, after H18)
Dataset	ds636
Phase selection	☐ Solution phase ☐ Pure phase
Diagram type	P-T diagram (fixed composition)
Solidus H ₂ O-saturated	false
TE predictive model	false
Pressure [kbar]	min: 0.01 max: 20.01
Temperature [°C]	min: 400 max: 1000
Initial grid subdivision	4 16 × 16 grid
Refinement type	Phases only
Refinement levels	4
Boost mode	true
Buffer	no buffer
Solver	Legacy
Verbose	none
Specific Cp	G0

- Method to compute specific heat capacity

$$Cp = -T \frac{\Delta^2 G}{\Delta T^2} = -T \frac{G^{(T+\delta T)} + G^{(T-\delta T)} - 2G^{(T)}}{\delta T^2}$$

where Cp is the specific heat capacity, T is the temperature

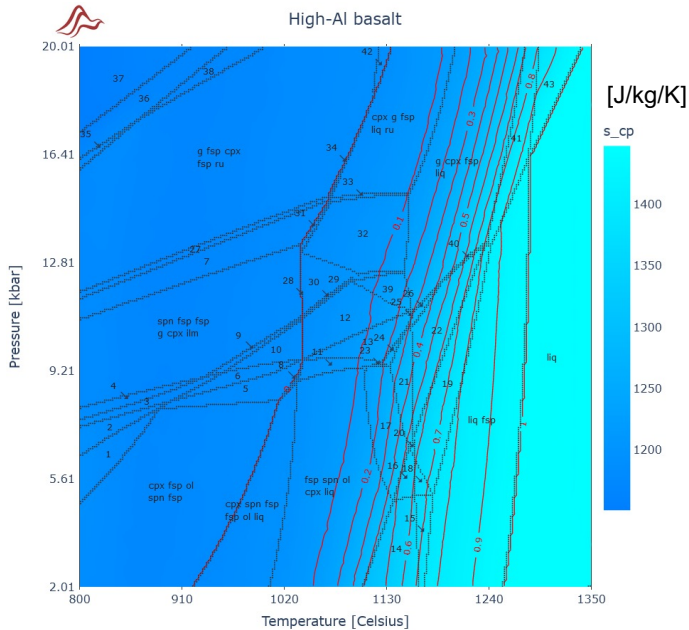
- Can be computed using

1. phase aggregate (G_0) **Without latent heat**
2. phase assemblage (G_{system}) **With latent heat**

- Note that when **G_system** is selected, solver is set to legacy
higher accuracy at suprasolidus conditions is needed to compute Gibbs second derivative

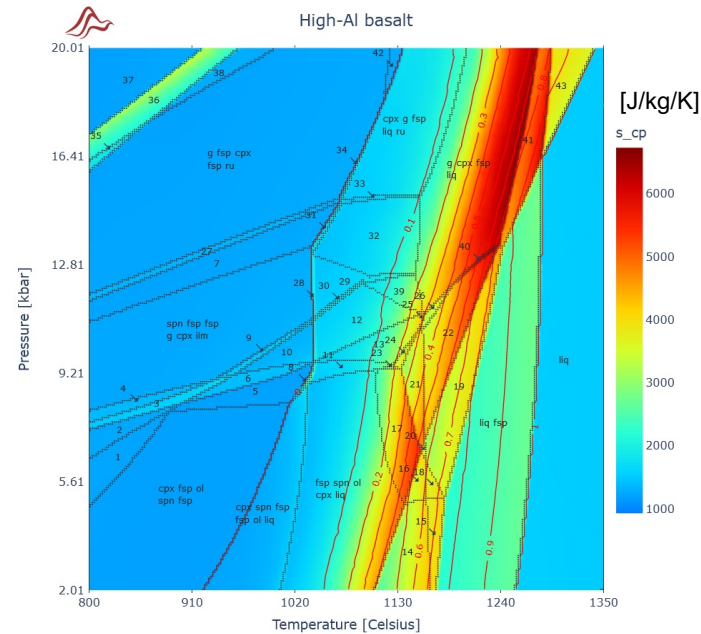
Specific heat capacity: what difference does it make?

G_0 differentiation
(Cp without latent heat of reaction)



- Cheap to compute
- Ignores latent heat

G_{system} differentiation
(Cp with latent heat of reaction)



- Slightly more expensive to compute
- Correct heat budget

MAGEMinApp: bulk rock composition

Bulk-rock composition

Drag and drop or select bulk-rock file

KLB1 Peridotite - Anhydrous

oxide	mol_fraction
SiO ₂	38.494
Al ₂ O ₃	1.776
CaO	2.824
MgO	50.566
FeO	5.886
K ₂ O	0.01
Na ₂ O	0.25
TiO ₂	0.1
O	0.096
Cr ₂ O ₃	0.109
H ₂ O	0

Buffer multiplier 0

ig

- Load custom bulk rock composition(s)
- Select predefined bulk rock composition for testing
- Bulk rock composition displayed in mol%
note that only the oxides taken into account in the database are displayed

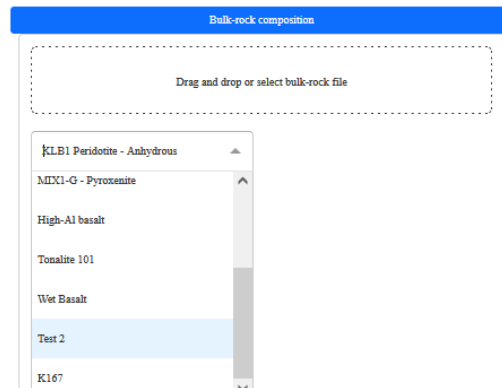
MAGEMinApp: bulk rock input file format

Example of input file provided in:

https://github.com/ComputationalThermodynamics/Resources/GG2024/bulkrock_composition_input_files/

```
# HEADER
title; comments; db; sysUnit; oxide; frac; frac2
# BULK-ROCK COMPOSITION
Test 2; Moo-et-al., 2000; ig; mol; [SiO2, Al2O3, CaO, MgO, FeO, K2O, Na2O, TiO2, O, Cr2O3, H2O]; [48.97, 11.76, 13.87, 4.21, 8.97, 1.66, 10.66, 1.36, 1.66, 0.01, 5.0];
K167; Moo-et-al., 2000; ig; mol; [SiO2, Al2O3, CaO, MgO, FeO, K2O, Na2O, TiO2, O, Cr2O3, H2O]; [50.3162, 8.0778, 11.6616, 11.3262, 9.1893, 0.4312, 2.4051, 2.271, 4.312, 0.0096, 0.0];
Test 3; Coin & Kwak, 1984; mb; wt; [SiO2, Al2O3, CaO, MgO, Fe2O3, K2O, Na2O, TiO2, FeO, H2O]; [55.12, 12.76, 4.32, 5.21, 2.45, 1.66, 10.66, 1.36, 1.66, 2.0];
Simon; Shorn-Simon, 2023; mb; mol; [SiO2, Al2O3, CaO, MgO, FeO, K2O, Na2O, TiO2, O, H2O]; [61.8, 8.57, 9.26, 5.05, 7.27, 0.21, 3.43, 1.25, 0.73, 2.85];
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- **Note that either [..., Fe2O3, FeO,...] or [..., FeO, O, ...] must be provided for the database accounting for O**



- **Once loaded, the bulk-rock composition appear in the dropdown menu**
Note that the bulk are attached to a database, this is why only Test 2 and K167 appear in the dropdown menu